

Robust Distributed Formation Control for Heterogeneous UAVs–UGVs Systems: An Adaptive Asymptotic Funnel Control Approach

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Abstract—This paper is concerned with the distributed formation control problem of heterogeneous uncrewed aerial vehicles–uncrewed ground vehicles (UAVs–UGVs) systems, in the presence of practical factors such as parametric and dynamics uncertainties, external disturbances, coupled dynamics, and communication noises. First, we establish a model for the heterogeneous UAVs–UGVs system with such practical factors. Subsequently, a new robust distributed formation control strategy is developed. To be specific, by implementing a delicate transformation with a special funnel function, together with fusing an adaptive asymptotic funnel control approach, a funnel-based formation control strategy is constructed. In comparison to the current outcomes, this strategy demonstrates the subsequent salient characteristics: 1) enhancing the robustness of the considered system subject to such practical factors; 2) achieving the asymptotic formation control with predefined global funnel performance constraints; 3) lowering computational and structural complexity without function approximation, switching of the controller structure, and complex nonlinear terms. Ultimately, the theoretical assertions are validated via the simulation paradigm.

Index Terms—Asymptotic stability, distributed formation control, funnel control, heterogeneous UAVs–UGVs systems, robust adaptive control.

I. INTRODUCTION

DISTRIBUTED formation control of heterogeneous multi-agent systems (MASs) has attracted significant attention

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due to its broad application potential [1], [2], [3]. Its primary goal is to stabilize the relative positions of heterogeneous MASs through local interactions, thereby establishing and maintaining a predefined geometric formation. As a representative case, the uncrewed aerial vehicles–uncrewed ground vehicles (UAVs–UGVs) system [4], comprising multiple UAVs and UGVs, significantly extends the search radius and service coverage by leveraging air-ground coordination and complementary functionalities [5]. Consequently, the heterogeneous UAVs–UGVs system enables tasks that are difficult for single-vehicle formations [6], [7] and is widely applied in 3D mapping [8], hybrid path planning [9], surveillance–reconnaissance mission [10], etc.

The distributed formation control problem for heterogeneous UAVs–UGVs systems has been thoroughly explored recently. Current advancements include a disturbance-resistant UAVs–UGVs formation [10] via distributed average tracking, a fault tolerant adaptive formation control framework [4], a human-in-the-loop 3D coordination [11] through unified models. However, in [4], [10], [11], the attitude dynamics of each UAV was ignored. Actually, the UAV must be maneuvered into a specific attitude to produce the required acceleration to achieve the target position [12]. Therefore, considering both the position and attitude dynamics of the UAV is essential. Additionally, in practical applications, the system parameters of the heterogeneous UAVs–UGVs system are prone to varying during operation [13] and inter-vehicle communication channels may suffer from uncertainties [14] such as packet loss [15], random quantization [16], and measurement noise [17]. These practical factors pose significant challenges for distributed formation control of heterogeneous UAVs–UGVs systems, particularly when precision in formation specification is required.

Defining the precision of formation specification often depends on the transient and steady-state values of the relative position errors between the neighboring vehicles or leader. To achieve a required precision, funnel control methods [18], [19], [20], as an effective tool, have been broadly utilized to ensure the system attains the desired transient and steady-state control performance. Alternatively, the prescribed performance control (PPC) approach [21], [22], [23] serves as another tool for preassigning both transient and steady-state behaviors.

Inspired by their nature, numerous funnel/PPC-based control methods were put forward for heterogeneous uncertain MASs. In [24], the event-triggered synchronization control issue of the heterogeneous uncertain UAVs-UGVs system with prescribed performance was studied, while neural networks were utilized in controller design. Similarly, several funnel/PPC control methods were constructed in [25], [26], [27] for heterogeneous uncertain MASs based on fuzzy-approximation approach. However, the employment of function approximation drastically increases the complexity of the controller [28], [29], [30].

Different from the function approximation-based funnel/PPC control methods, the low-complexity funnel/PPC control without using function approximation has been widely investigated. To mention a few, [29] and [30] with relative-degree-two MASs, [31] with marine surface vessels, and [32], [33] with mismatched uncertainties MASs. Nevertheless, two primary concerns still require attention. One concern is that the feasibility of these control laws depends on local, known initial conditions, which is a restrictive requirement for heterogeneous UAVs-UGVs systems where platform heterogeneity inherently leads to distributed, unknown initial states. While recent advances [34], [35], [36] progressively relax these assumptions for SISO systems, their reliance on homogeneous state dimensions renders them inapplicable to cross-domain systems where aerial and ground vehicles exhibit fundamentally different dynamic orders. The second concern is that merely confining the concerned error within a predefined funnel, which leads to a bounded steady-state error (referred to as “practical stability”), is insufficient for precision tasks such as cooperative payload transport that require asymptotic convergence. More recently, non-smooth feedback approaches [37], [38], [39] achieve asymptotic tracking but at the cost of structural complexity incompatible with the computational constraints of heterogeneous UAVs-UGVs systems. To sum up, these two unresolved challenges motivate our investigation into smooth asymptotic global funnel control strategy in a low-complexity form for heterogeneous UAVs-UGVs systems. Specifically, the integration of such control solutions onto heterogeneous UAVs-UGVs systems necessitates controller designs with inherently low computational complexity, while maintaining the cross-domain coordination requirements of global initial-condition independence and asymptotic precision.

Motivated by the preceding observations, we construct a new adaptive asymptotic funnel control approach for designing a robust distributed formation control strategy for heterogeneous UAVs-UGVs systems. Specifically, the major innovations of this study are outlined below.

- 1) A fairly general class of heterogeneous UAVs-UGVs systems with communication noises and dynamics uncertainties is considered, where both the position and attitude dynamics of UAVs are simultaneously involved.
- 2) By applying a nonlinear function together with a specific funnel function, a clever transformation is performed such that the initial error consistently remains within a non-empty open set. Building upon this transformation, an adaptive asymptotic funnel control approach is developed by integrating the bound estimation method

with funnel control. This approach guarantees asymptotic formation control while satisfying arbitrarily predefined global funnel performance constraints, in contrast to the current funnel/PPC methods in [29], [30], [31], [32], [33], which require known initial conditions and merely attain “practical stability.”

- 3) Compared with the existing funnel-based asymptotic results [37], [38], [39], the proposed strategy eliminates the need for controller structure switching by introducing compensating terms in the form of smooth functions with respect to formation errors and certain Lebesgue integrable functions. Furthermore, it incorporates fewer non-linear terms, which is achieved by leveraging the adaptive law’s inherent properties with suitable scaling.

The remainder is structured as follows. Section II presents the mathematical preliminaries and problem formulation. Section III shows the main results, containing the proposal of a robust distributed formation control and its stability analysis. The theoretical outcomes are demonstrated using numerical simulations in Section IV, and the entire paper is concluded in Section V.

II. PRELIMINARIES AND PROBLEM FORMULATION

A. Notation

$\mathbf{R}$, $\mathbf{R}^N$, and $\mathbf{R}^{N \times n}$, and $\mathbf{R}_+$ denote the sets of real numbers, real vectors of order N , real matrices of size $N \times n$, and positive real numbers, respectively. $\mathbf{1}_N \in \mathbf{R}^N$ represents a vector of ones, while $\mathbf{I}_N$ signifies the identity matrix of size $N \times N$. $\mathbf{N}_n^1$ stands for $\{1, \dots, n\}$. $\|\cdot\|$ indicates the Euclidean norm of a vector or the induced 2-norm of a matrix. $\mathbf{S}(\mathbf{x}) \in \mathbf{R}^{n \times n}$ represents a skew-symmetric matrix corresponding to $\mathbf{x} \in \mathbf{R}^n$. The special orthogonal group is denoted by $\mathbf{SO}(n) := \{\mathbf{R} \in \mathbf{R}^{n \times n} : \mathbf{R}\mathbf{R}^T = \mathbf{I}_n, \det(\mathbf{R}) = 1\}$. $\mathcal{L}_\infty$ represent the space of essentially bounded. Lebesgue integrable functions and square integrable functions are denoted by $\mathcal{L}_1$ and $\mathcal{L}_2$.

B. Graph Theory

Let’s use a digraph $\overline{\mathcal{G}} = (\overline{\mathcal{V}}, \overline{\mathcal{E}})$ to represent the interaction among $N_a (\geq 1)$ UAV(s) and $N_g (\geq 1)$ UGV(s) and one (virtual) leader, and use the subgraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ to describe the communication graph of $N (= N_a + N_g)$ vehicles, where the set of vehicles $\mathcal{V} = \overline{\mathcal{V}} \setminus \{0\}$ with $\overline{\mathcal{V}} = \{1, \dots, N, 0\}$, and the set of edges $\overline{\mathcal{E}} \subseteq \overline{\mathcal{V}}^2$ and $\mathcal{E} \subseteq \mathcal{V}^2$. It is assumed that all UAVs are in the group $\mathbb{M}_a$ while the remaining UGVs are grouped into group $\mathbb{M}_g$. The neighbors of the i -th vehicle are denoted as $\tilde{\mathcal{N}}_i = \{l : (l, i) \in \overline{\mathcal{E}}\}$ and $\mathcal{N}_i = \{l : (l, i) \in \mathcal{E}\}$. The adjacency matrices of $\overline{\mathcal{G}}$ and $\mathcal{G}$ are given as $\overline{\mathbf{A}}$ and $\mathbf{A} = [a_{il}]_{N \times N}$ with $a_{il} = 1$ if $l \in \tilde{\mathcal{N}}_i$, and $a_{il} = 0$, otherwise. $\overline{\mathbf{A}}$ is defined as $\overline{\mathbf{A}} = [\mathbf{A}, \mathbf{b}_0; \mathbf{0}_{1 \times N}, 0] \in \mathbf{R}^{(N+1) \times (N+1)}$, where $\mathbf{b}_0 = [b_1, \dots, b_N]^T$ with $b_i = 1$ if $0 \in \tilde{\mathcal{N}}_i$, otherwise $b_i = 0$. Denote the Laplacian matrix of $\overline{\mathcal{G}}$ and $\mathcal{G}$ as $\overline{\mathbf{L}}$ and $\mathbf{L} = [\mathbb{L}_{il}]_{N \times N}$ with $\mathbb{L}_{il} = -a_{il}$ for $i \neq l, i, l \in \mathcal{V}$, and $\mathbb{L}_{ii} = \sum_{l \in \tilde{\mathcal{N}}_i} a_{il}$, respectively. $\overline{\mathbf{L}}$ is defined as $\overline{\mathbf{L}} = [\mathbf{H}, -\mathbf{b}_0; \mathbf{0}_{1 \times N}, 0]_{(N+1) \times (N+1)}$ with $\mathbf{H} = \mathbf{L} + \mathbf{B}$ and $\mathbf{B} = \text{diag}\{b_1, \dots, b_N\}$.

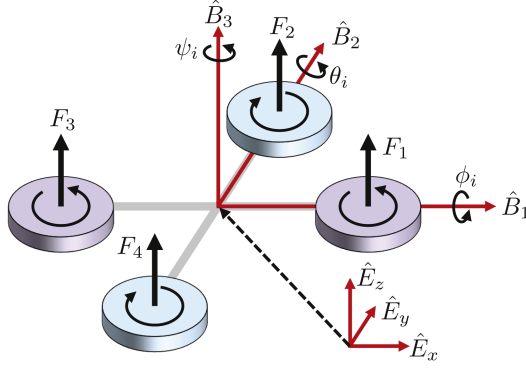


Fig. 1. Configuration of the i -th UAV: $\{\hat{E}_x, \hat{E}_y, \hat{E}_z\}$ and $\{\hat{B}_1, \hat{B}_2, \hat{B}_3\}$ denote the earth frame and body frame.

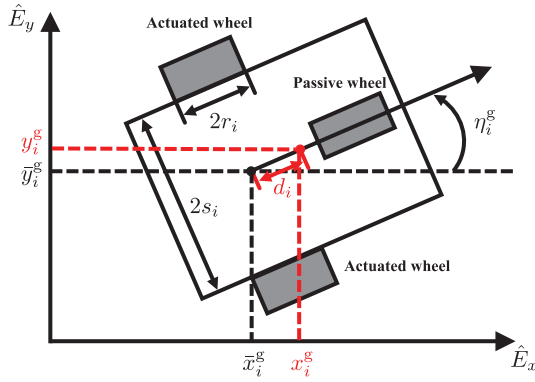


Fig. 2. Configuration of the i -th UGV: $\{\hat{E}_x, \hat{E}_y\}$ represents the earth frame.

C. Modeling of the Heterogeneous UAVs-UGVs System

1) *UAV Dynamics:* The configuration of $i \in \mathbb{M}_a$ -th UAV is shown in Fig. 1. Define these two frames as $\hat{E} = \{\hat{E}_x, \hat{E}_y, \hat{E}_z\}$ and $\hat{B} = \{\hat{B}_1, \hat{B}_2, \hat{B}_3\}$, the position of the i -th UAV in $\hat{E}$ as $\mathbf{p}_i^a = [p_{i,x}^a, p_{i,y}^a, p_{i,z}^a]^T \in \mathbb{R}^3$, and its Euler angles as $\boldsymbol{\eta}_i^a = [\phi_i, \theta_i, \psi_i]^T \in \mathbb{R}^3$, and then the dynamics of the i -th UAV is expressed as [12]

$$\begin{aligned} m_i^a \ddot{\mathbf{p}}_i^a &= \mathfrak{R}^{\text{EB}}(\boldsymbol{\eta}_i^a) \mathbf{f}_i^a + \boldsymbol{\vartheta}_{p,i}^a(t), \\ \mathcal{I}_i^a \ddot{\boldsymbol{\eta}}_i^a &= -\mathbf{S}(\boldsymbol{\eta}_i^a) \mathcal{I}_i^a \dot{\boldsymbol{\eta}}_i^a + \boldsymbol{\tau}_i^a + \boldsymbol{\vartheta}_{\eta,i}^a(t), \end{aligned} \quad (1)$$

where $m_i^a > 0$ represents the i -th UAV's mass, $\mathfrak{R}^{\text{EB}}(\boldsymbol{\eta}_i^a) \in \text{SO}(3)$ represents the rotation matrix bringing $\hat{B}$ to $\hat{E}$, which can be found in [40], $\mathcal{I}_i^a = \text{diag}(\mathcal{I}_{\phi,i}^a, \mathcal{I}_{\theta,i}^a, \mathcal{I}_{\psi,i}^a) \in \mathbb{R}^{3 \times 3}$ is the inertia matrix, and $\mathbf{f}_i^a = k_{\omega,i} \mathbf{u}_{z,i}^a \mathbf{o}_3 - (\mathfrak{R}^{\text{EB}}(\boldsymbol{\eta}_i^a))^T \mathbf{o}_3 m_i^a g \in \mathbb{R}^3$ and $\boldsymbol{\tau}_i^a = [k_{\omega,i} l_{c,i} u_{\phi,i}, k_{\omega,i} l_{c,i} u_{\theta,i}, k_{\tau,i} u_{\psi,i}]^T \in \mathbb{R}^3$ are respectively denote the external forces and torques with $\mathbf{o}_3 = [0, 0, 1]^T$, the gravity g , and the control inputs $u_{z,i} = \sum_{q=1}^4 \omega_{j,i}^2$, $u_{\phi,i} = \omega_{2,i}^2 - \omega_{4,i}^2$, $u_{\theta,i} = \omega_{3,i}^2 - \omega_{1,i}^2$, and $u_{\psi,i} = \sum_{q=1}^4 (-1)^{j+1} \omega_{j,i}^2$, where $\omega_{j,i}$ denotes the j -th rotor speed, and $l_{c,i}, k_{\omega,i}, k_{\tau,i} > 0$ are the scaling factors, $[(\boldsymbol{\vartheta}_{p,i}^a(t))^T, (\boldsymbol{\vartheta}_{\eta,i}^a(t))^T]^T \in \mathbb{R}^6$ denotes the disturbances.

2) *UGV Dynamics:* As illustrated in Fig. 2, the mathematical model of the $i \in \mathbb{M}_g$ -th UGV can be modeled as [4]

$$\begin{aligned} \begin{bmatrix} \dot{\mathbf{p}}_i^g \\ \dot{\boldsymbol{\eta}}_i^g \end{bmatrix} &= \begin{bmatrix} \cos(\eta_i^g) & 0 \\ \sin(\eta_i^g) & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_i^g \\ \omega_i^g \end{bmatrix}, \\ \begin{bmatrix} \dot{v}_i^g \\ \dot{\omega}_i^g \end{bmatrix} &= \begin{bmatrix} \frac{1}{m_i^g r_i} & \frac{1}{m_i^g r_i} \\ \frac{s_i}{\mathcal{I}_i^g r_i} & -\frac{s_i}{\mathcal{I}_i^g r_i} \end{bmatrix} \boldsymbol{\tau}_i^g + \boldsymbol{\vartheta}_i^g(t), \end{aligned} \quad (2)$$

where $\bar{\mathbf{p}}_i^g(t) = [\bar{x}_i^g(t), \bar{y}_i^g(t)]^T \in \mathbb{R}^2$ is the i -th UGV's position in $\{\hat{E}_x, \hat{E}_y\}$, $\eta_i^g(t)$ denotes its heading angle, $v_i^g(t)$ and $\omega_i^g(t)$ are the linear and angular velocities, respectively; the radius of the wheel and half width of the i -th UGV are denoted by r_i and s_i , respectively; $\mathcal{I}_i^g$ and m_i^g indicate the moment of inertia and mass of the i -th UGV, respectively, $\boldsymbol{\tau}_i^g \in \mathbb{R}^2$ represents the control torque, the disturbance is denoted by $\boldsymbol{\vartheta}_i^g(t) \in \mathbb{R}^2$.

From observing (1) and (2), it is evident that the number of inputs for the UAV (i.e., $u_{z,i}$, $u_{\phi,i}$, $u_{\theta,i}$, and $u_{\psi,i}$) and UGV (i.e., $\boldsymbol{\tau}_i^g \in \mathbb{R}^2$) is fewer than the number of configuration variables for the UAV (i.e., $\mathbf{p}_i^a \in \mathbb{R}^3$ and $\boldsymbol{\eta}_i^a \in \mathbb{R}^3$) and UGV (i.e., $\bar{\mathbf{p}}_i^g \in \mathbb{R}^2$ and η_i^g). Therefore, the dynamics of both UAV and UGV are under-actuated systems. To this end, we formulate a virtual position control input

$$\mathbf{u}_{p,i}^a = u_{z,i}^a \mathfrak{R}^{\text{EB}}(\boldsymbol{\eta}_{r,i}^a) \mathbf{o}_3 \in \mathbb{R}^3, \quad (3)$$

with $\boldsymbol{\eta}_{r,i}^a = [\phi_{r,i}, \theta_{r,i}, \psi_{r,i}]^T$ being the reference attitude for the i -th UAV, which requires to be tracked through the controller of attitude system. Noted that the extraction of the desired attitude is required since the virtual position controller is not directly force-generated. Using the fact that $\|\mathfrak{R}^{\text{EB}}(\boldsymbol{\eta}_{r,i}^a) \mathbf{o}_3\| = 1$, as the reference of yaw angle $\psi_{r,i}$ is preset, the control input $u_{z,i}^a$ along with the reference of pitch angle and the roll angle are derived as $u_{z,i}^a = \|\mathbf{u}_{p,i}^a\|$, $\phi_{r,i} = \sin^{-1}((u_{i,1} \sin(\psi_{r,i}) - u_{i,2} \cos(\psi_{r,i})) / u_{z,i}^a)$, and $\theta_{r,i} = \tan^{-1}((u_{i,1} \cos(\psi_{r,i}) - u_{i,2} \sin(\psi_{r,i})) / u_{i,3})$.

By utilizing $\mathbf{u}_{p,i}^a$, the first equation of (1) can be rewritten as $m_i^a \ddot{\mathbf{p}}_i^a = k_{\omega,i} \mathbf{u}_{p,i}^a - k_{\omega,i} \mathbf{u}_{p,i}^a + \mathfrak{R}^{\text{EB}}(\boldsymbol{\eta}_i^a) \mathbf{f}_i^a + \boldsymbol{\vartheta}_{p,i}^a(t)$. Then, consider the parameter uncertainty, by defining $B_{p,i}^a = (m_i^a)^{-1} k_{\omega,i} \mathbf{I}_3 = B_{p,i}^{a,N} + B_{p,i}^{a,\Delta}$ and $B_{\eta,i}^a = (\mathcal{I}_i^a)^{-1} \text{diag}(k_{\omega,i} l_{c,i}, k_{\omega,i} l_{c,i}, k_{\tau,i}) = B_{\eta,i}^{a,N} + B_{\eta,i}^{a,\Delta}$ with the subscripts N and Δ denoting the nominal parameter and the parameter uncertainty, the model (1) can be reformulated as

$$\begin{aligned} \ddot{\mathbf{p}}_i^a &= B_{p,i}^{a,N} \mathbf{u}_{p,i}^a - g \mathbf{o}_3 + B_{p,i}^{a,N} \tilde{\mathbf{u}}_{p,i}^a + (m_i^a)^{-1} \boldsymbol{\vartheta}_{p,i}^a(t), \\ \ddot{\boldsymbol{\eta}}_i^a &= B_{\eta,i}^{a,N} \mathbf{u}_{\eta,i}^a - (\mathcal{I}_i^a)^{-1} \mathbf{S}(\boldsymbol{\eta}_i^a) \mathcal{I}_i^a \dot{\boldsymbol{\eta}}_i^a + \Delta_{\eta,i}^a, \end{aligned} \quad (4)$$

with $\tilde{\mathbf{u}}_{p,i}^a = u_{z,i}^a (B_{p,i}^{a,N})^{-1} B_{p,i}^{a,\Delta} \mathfrak{R}^{\text{EB}}(\boldsymbol{\eta}_i^a) \mathbf{o}_3 - \mathbf{u}_{p,i}^a$ being identified as the coupled term arising from the discrepancy in attitude tracking, $\mathbf{u}_{\eta,i}^a = [u_{\phi,i}, u_{\theta,i}, u_{\psi,i}]^T \in \mathbb{R}^3$ representing the control torques, $\Delta_{\eta,i}^a = B_{\eta,i}^{a,\Delta} \mathbf{u}_{\eta,i}^a + (\mathcal{I}_i^a)^{-1} \boldsymbol{\vartheta}_{\eta,i}^a(t)$.

Then, for the i -th UGV, an onboard sensor shown in Fig. 2 is installed at the position $\mathbf{p}_i^g(t) = [x_i^g(t), y_i^g(t)]^T \in \mathbb{R}^2$, which is offset by a distance $d_i > 0$ from $\bar{\mathbf{p}}_i^g(t)$ in the direction specified by $\eta_i^g(t)$. The position $\mathbf{p}_i^g(t)$ is described by $\mathbf{p}_i^g(t) = \bar{\mathbf{p}}_i^g(t) +$

$d_i[\cos(\eta_i^g(t)), \sin(\eta_i^g(t))]^T$, whose dynamics is modeled as

$$\ddot{\mathbf{p}}_i^g = \begin{bmatrix} -v_i^g \omega_i^g \sin(\eta_i^g) - d_i(\omega_i^g)^2 \cos(\eta_i^g) \\ v_i^g \omega_i^g \cos(\eta_i^g) - d_i(\omega_i^g)^2 \sin(\eta_i^g) \end{bmatrix} + \underbrace{\begin{bmatrix} \cos(\eta_i^g) & -d_i \sin(\eta_i^g) \\ \sin(\eta_i^g) & d_i \cos(\eta_i^g) \end{bmatrix} \begin{bmatrix} \frac{1}{m_i^g r_i} & \frac{1}{m_i^g r_i} \\ \frac{s_i^g}{I_i^g r_i} & -\frac{s_i^g}{I_i^g r_i} \end{bmatrix}}_{B_i^g} \boldsymbol{\tau}_i^g. \quad (5)$$

By defining $B_i^g = B_i^{g,N} + B_i^{g,\Delta}$, which yields

$$\ddot{\mathbf{p}}_i^g = B_i^{g,N} \boldsymbol{\tau}_i^g + \mathbf{f}_i^g + \Delta_i^g, \quad (6)$$

where $\mathbf{f}_i^g = [-v_i^g \omega_i^g \sin(\eta_i^g) - d_i(\omega_i^g)^2 \cos(\eta_i^g), v_i^g \omega_i^g \cos(\eta_i^g) - d_i(\omega_i^g)^2 \sin(\eta_i^g)]^T$ and $\Delta_i^g = B_i^{g,\Delta} \boldsymbol{\tau}_{\eta,i}^g + \boldsymbol{\vartheta}_i^g(t)$. It is noted that $B_i^{g,N}$ is non-singular since $\det(B_i^{g,N}) \neq 0$.

Thus, the heterogeneous UAVs-UGVs system is modeled as

$$\begin{aligned} \forall i \in \mathbb{M}_a : \begin{cases} \dot{\mathbf{p}}_i^a = B_{p,i}^{a,N} \mathbf{u}_{p,i}^a - g\mathbf{o}_3 + \Delta_{p,i}^a, \\ \dot{\boldsymbol{\eta}}_i^a = B_{\eta,i}^{a,N} \mathbf{u}_{\eta,i}^a - (\mathcal{I}_i^a)^{-1} \mathbf{S}(\boldsymbol{\eta}_i^a) \mathcal{I}_i^a \boldsymbol{\eta}_i^a + \Delta_{\eta,i}^a, \end{cases} \\ \forall i \in \mathbb{M}_g : \begin{cases} \ddot{\mathbf{p}}_i^g = B_i^{g,N} \boldsymbol{\tau}_i^g + \mathbf{f}_i^g + \Delta_i^g, \\ \dot{\boldsymbol{\eta}}_i^g = \omega_i^g, \dot{\omega}_i^g = \begin{bmatrix} \frac{s_i^g}{I_i^g r_i}, -\frac{s_i^g}{I_i^g r_i} \end{bmatrix} \boldsymbol{\tau}_i^g + \boldsymbol{\vartheta}_{i,2}^g(t), \end{cases} \end{aligned} \quad (7)$$

where $\Delta_{p,i}^a = B_{p,i}^{a,N} \tilde{\mathbf{u}}_{p,i}^a + (m_i^a)^{-1} \boldsymbol{\vartheta}_{p,i}^a(t)$.

Remark 1: Unlike the related literature [4], [10], [11], the established model is more general for the following reasons: 1) it simultaneously considers the dynamics of both UAV position and attitude, including the coupling effects present in attitude tracking; 2) uncertain external disturbances are taken into account in both position and attitude; 3) system parameter uncertainty is considered, and the inherent nonlinear functions are unknown smooth functions.

D. Problem Formulation

This paper focus on the distributed formation control problem of the heterogeneous UAVs-UGVs system (7). To simplify notation, we introduce the variable $\mathbf{p}_i \in \mathbb{R}^3$ defined by $\mathbf{p}_i = \mathbf{p}_i^a$ if $i \in \mathbb{M}_a$; otherwise, $\mathbf{p}_i = [(\mathbf{p}_i^g)^T, 0]^T$. First, the target formation configuration $\mathbb{F}_t$ is specified according to the relative positions of the vehicles as follows:

$$\mathbb{F}_t := \{(\mathbf{p}, \mathbf{p}_0) \in \mathbf{R}^{3N} \times \mathbf{R}^3 : \mathbf{p}_i - \mathbf{p}_l = \boldsymbol{\zeta}_{il}, (l, i) \in \overline{\mathbb{E}}\}, \quad (8)$$

where $\mathbf{p} = [\mathbf{p}_1^T, \dots, \mathbf{p}_N^T]^T \in \mathbf{R}^{3N}$, $\boldsymbol{\zeta}_{il} = [\zeta_{il,1}, \zeta_{il,2}, \zeta_{il,3}]^T \in \mathbf{R}^3$ denotes the desired position deviation between the i -th and l -th vehicles. The deviation $\boldsymbol{\zeta}_{il}$ is calculated according to $\boldsymbol{\zeta}_{il} = \boldsymbol{\zeta}_{i0} - \boldsymbol{\zeta}_{l0}$, where $\boldsymbol{\zeta}_{i0} = [\zeta_{i0,1}, \zeta_{i0,2}, \zeta_{i0,3}]^T \in \mathbf{R}^3$ represents the desired position deviation between the leader and the i -th vehicle. Denote $\mathbf{p}_0 = [p_{0,x}, p_{0,y}, p_{0,z}]^T \in \mathbf{R}^3$ as the leader's position in $\hat{E}_I$, whose dynamics is assumed as $\dot{\mathbf{p}}_0 = \mathbf{f}_0(\mathbf{p}_0, t)$ with $\mathbf{f}_0 : \mathbf{R}^3 \times [0, \infty) \rightarrow \mathbf{R}^3$ being locally Lipschitz on $\mathbf{p}_0$.

Based on the aforementioned description, the formation error of the i -th vehicle is defined as:

$$\mathbf{e}_{i,1} = \sum_{l \in \mathbb{N}_i} a_{il}(\mathbf{p}_i - \mathbf{p}_l - \boldsymbol{\zeta}_{il}), \quad i \in \mathbb{V}. \quad (9)$$

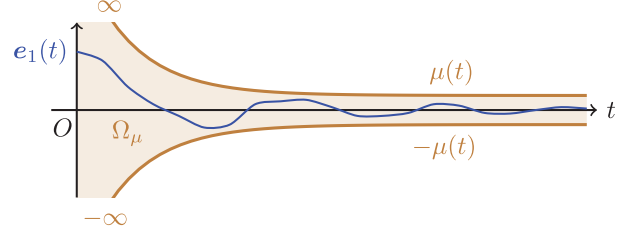


Fig. 3. Schematic illustration of the evolution of formation error inside the funnel (Projection in a certain dimension).

It is noticed that, in the context of distributed formation control involving multiple UAVs and UGVs, each vehicle can merely obtain restricted local information regarding itself and its neighbors. As a result, $\boldsymbol{\zeta}_{i0}$ cannot be accessed if the i -th vehicle does not have a connection to the leader.

In many realistic scenarios, the communication noises may exist in the communication channels. Similar to [17], we consider a transmission system affected by multiplicative uncertainty. Namely, the information transmitted from l to i at time t consists in the relative position $\mathbf{p}_i(t) - \mathbf{p}_l(t) - \boldsymbol{\zeta}_{il}$ up to a multiplication by $\check{a}_{il} := a_{il} + \bar{a}_{il}(t)$, i.e., $\check{a}_{il}(t)(\mathbf{p}_i(t) - \mathbf{p}_l(t) - \boldsymbol{\zeta}_{il})$, where $\bar{a}_{il}(t)$ is an unknown perturbed function. Thus, the formation error of the i -th vehicle should be reformulated as $\mathbf{e}_{i,1} = \sum_{l \in \mathbb{N}_i} \check{a}_{il}(t)(\mathbf{p}_i - \mathbf{p}_l - \boldsymbol{\zeta}_{il})$. Denote $\mathbf{e}_1 = [\mathbf{e}_{1,1}^T, \dots, \mathbf{e}_{N,1}^T]^T$, and $\boldsymbol{\zeta} = [\boldsymbol{\zeta}_{10}^T, \dots, \boldsymbol{\zeta}_{N0}^T]^T$, it can be compactly written as

$$\mathbf{e}_1(t) = \mathbb{H}(t)(\mathbf{p} - \mathbf{1}_N \otimes \mathbf{p}_0 - \boldsymbol{\zeta}) = \mathbb{H}(t)\mathbf{E}, \quad (10)$$

where $\mathbb{H}(t)$ replaces $\mathbb{H}$ due to the presence of $\bar{a}_{il}(t)$, $\mathbf{E} = [\mathbf{E}_1^T, \dots, \mathbf{E}_N^T]^T := \mathbf{p} - \mathbf{1}_N \otimes \mathbf{p}_0 - \boldsymbol{\zeta}$ denotes the position tracking error.

As shown in Fig. 3, the control objective is to design a distributed control strategy such that

- 1) the formation error $\mathbf{e}_1(t)$ always evolves within the following performance funnel

$$\Omega_\mu = \{(\mathbf{e}_1(t), t) \in \mathbf{R}^{3N} \times [0, \infty) : \|\mathbf{e}_1(t)\| < \mu(t)\}, \quad (11)$$

in which $\mu(t)$ signifies a funnel function, which is continuous and differentiable for all t , $\sup_{t \geq 0} 1/\mu(t) \geq 0$, $\sup_{t \geq 0} 1/\mu(t) < \infty$, and $\mu(0) = \infty$;

- 2) $\mathbf{e}_1(t)$ reaches zero as $t \rightarrow \infty$, which signifies that the target formation configuration $\mathbb{F}_t$ is achieved, i.e., $\lim_{t \rightarrow \infty} (\mathbf{p}_i - \mathbf{p}_l) = \boldsymbol{\zeta}_{il}$, for all $(l, i) \in \overline{\mathbb{E}}$.

To achieve such objective, the following standard assumptions are required.

Assumption 1: A spanning tree exists in $\mathbb{G}$, and the root vehicle i_r has access to information from the leader, namely, $b_{i_r} > 0$.

Assumption 2: The continuous perturbed function $\bar{a}_{il}(t)$ and its derivative $\dot{\bar{a}}_{il}(t)$ fulfill $\bar{a}_{il}(t), \dot{\bar{a}}_{il}(t) \in \mathcal{L}_\infty$, with the signs of $\check{a}_{il}(t)$ being identical to those of a_{il} .

Assumption 3: The uncertainties which encompass both parameter uncertainties and external disturbances, are bounded and continuous.

Assumption 4: The leader's position p_0 is bounded and its dynamics is unknown but bounded. Additionally, the reference of yaw angle $\psi_{r,i}$ and its time derivative are bounded.

Remark 2: Assumptions 1–2 are standard conditions [17] that enable the realization of the formation control under communication noises. As described in [17], this perturbed communication channel can include a broad category of communication noises, e.g., quantization and signal distortion. Assumption 3 implies that the energy of external interference is limited, can be founded in [13]. Assumption 4 is a standard assumption [35], [36].

Remark 3: The control objective of this study is technically challenging, as it requires not only asymptotic convergence of the error to zero but also keeping it constrained within a predetermined funnel region, regardless of the uncertainty in initial conditions. Moreover, the initial value and dynamics of the leader's signal are not *prior* known, and the system may also suffer from communication noises, preventing the direct application of the correct information $p_i - p_l - \zeta_{il}$, $l \in \bar{N}_i$.

III. MAIN RESULTS

In this part, a robust distributed formation control strategy is developed by utilizing the adaptive asymptotic funnel control approach, which can ensure that vehicles can achieve formation configuration under unknown communication noises. Before the control design, we employ a funnel function $\mu_{i,k,q} : [0, \infty) \rightarrow \mathbf{R}_+$, $i \in \mathbb{V}$, $k \in \mathbf{N}_1^2$, $q \in \mathbf{N}_1^{n_i}$ as [19]

$$\mu_{i,k,q}(t) = (\mu_{i,k,q}^0 - \mu_{i,k,q}^\infty) e^{-\frac{\mu_{i,k,q}^*}{\mu_{i,k,q}^0} t} + \mu_{i,k,q}^\infty, \quad (12)$$

where $\mu_{i,k,q}^0, \mu_{i,k,q}^*, \mu_{i,k,q}^\infty > 0$, which satisfies $\mu_{i,k,q}(t) \in (0, \mu_{i,k,q}^0]$ and $|\dot{\mu}_{i,k,q}(t)| < \mu_{i,k,q}^*$; $n_i = 3$ if $i \in \mathbb{M}_a$ and $n_i = 2$ if $i \in \mathbb{M}_g$.

A. Robust Distributed Formation Control Design

The design process is executed through two parts.

Part I (Attitude Controller Design ($i \in \mathbb{M}_a$)): Select the funnel function $\mu_{i,1,q}^a(\cdot)$ in the form of (12), ensuring that $\mu_{i,1,q}^a(0) = 1$ and $\mu_{i,1,q}^a = h_{i,1,q}(o_{i,1,q}^a)$, where $0 < o_{i,1,q}^a < 1$, and $h_{i,1,q}(\cdot)$ is defined later. First, we define $e_{i,1,q}^a = \eta_i^a - \eta_{r,i}^a$, based on the element-wise multi-dimensional transformation, the virtual control law $\nu_i^a(t) = [\nu_{i,1}^a(t), \dots, \nu_{i,n_i}^a(t)]^T$ is constructed as

$$\nu_{i,q}^a = -c_{i,1}^a \epsilon_{i,1,q}^a - \hat{\varrho}_{i,1,q}^a \sigma_{i,1,q}^a(\epsilon_{i,1,q}^a), \quad q \in \mathbf{N}_1^3, \quad (13)$$

$$\hat{\varrho}_{i,1,q}^a = \gamma_{i,1}^a |\epsilon_{i,1,q}^a| - \bar{\gamma}_{i,1}^a \varpi_{i,1}(t) \hat{\varrho}_{i,1,q}^a, \quad \hat{\varrho}_{i,1,q}^a(0) \geq 0, \quad (14)$$

where $c_{i,1}^a, \gamma_{i,1}^a$, and $\bar{\gamma}_{i,1}^a$ are positive design parameters; the transformed error $\epsilon_{i,1,q}^a : (-1, 1) \rightarrow \mathbf{R}$, $\epsilon_{i,1,q}^a(0) = 0$ is given as $\epsilon_{i,1,q}^a(\check{e}_{i,1,q}^a) = \log((1 + \check{e}_{i,1,q}^a)/(1 - \check{e}_{i,1,q}^a))$ where $\check{e}_{i,1,q}^a = h_{i,1,q}(e_{i,1,q}^a)/\mu_{i,1,q}^a(t)$ with a nonlinear function

$$h_{i,1,q}(e_{i,1,q}^a) = \frac{\alpha_{i,1}^{2e_{i,1,q}^a} - 1}{\alpha_{i,1}^{2e_{i,1,q}^a} + 1}, \quad \alpha_{i,1} > 1, \quad (15)$$

where $h_{i,1,q}(0) = 0$ and $e_{i,1,q}^a$ is the j -element of $e_{i,1}^a$; the term $-\gamma_{i,1} \varpi_{i,1}(t) \hat{\varrho}_{i,1,q}^a$ is employed to preventing the parameter estimation drifting [41]; $\varpi_{i,k} : [0, \infty) \rightarrow \mathbf{R}_+ \in \mathcal{L}_1$ satisfies $\int_0^t \varpi_{i,k}(\tau) d\tau \leq \varpi_{i,k}^*$ with $\varpi_{i,k}^*$ being positive constants, for $k \in \mathbf{N}_1^2$, $\hat{\varrho}_{i,1,q}^a(t)$ is an adaptive gain, and

$$\sigma_{i,1,q}^a(\epsilon_{i,1,q}^a) = \frac{\hat{\varrho}_{i,1,q}^a \epsilon_{i,1,q}^a}{\sqrt{|\hat{\varrho}_{i,1,q}^a \epsilon_{i,1,q}^a|^2 + \varpi_{i,1}^2(t)}}. \quad (16)$$

Then, define $e_{i,2}^a(t) = \dot{\eta}_{i,2}(t) - \nu_i^a(t)$ and choose an auxiliary funnel function $\mu_{i,2,q}^a(\cdot)$ so that $\mu_{i,2,q}^a(0) = 1$ and $\mu_{i,2,q}^a = h_{i,2,q}(o_{i,2,q}^a)$, where $0 < o_{i,2,q}^a < 1$ and $h_{i,2,q}(\cdot)$ is defined later. Likewise, the actual control signal $\mathbf{u}_{\eta,i}^a(t) = [u_{i,1}^a, \dots, u_{i,n_i}^a]^T$ for the attitude system is constructed as

$$u_{i,q}^a = \sum_{k=1}^3 [B_{\eta,i}^{a,N}]_{q,k} (-c_{i,2}^a \epsilon_{i,2,k}^a - \hat{\varrho}_{i,2,k}^a \sigma_{i,2,k}^a(\epsilon_{i,2,k}^a)), \quad (17)$$

$$\dot{\varrho}_{i,2,q}^a = \gamma_{i,2}^a |\epsilon_{i,2,q}^a| - \bar{\gamma}_{i,2}^a \varpi_{i,2}(t) \hat{\varrho}_{i,2,q}^a, \quad \hat{\varrho}_{i,2,q}^a(0) \geq 0, \quad (18)$$

in which $c_{i,2}^a, \gamma_{i,2}^a$, and $\bar{\gamma}_{i,2}^a$ are positive design parameters, $\epsilon_{i,2,q}^a(\check{e}_{i,2,q}^a) = \log((1 + \check{e}_{i,2,q}^a)/(1 - \check{e}_{i,2,q}^a))$ with $\check{e}_{i,2,q}^a = h_{i,2,q}(e_{i,2,q}^a)/\mu_{i,2,q}^a(t)$ and $h_{i,2,q}(e_{i,2,q}^a) = (\alpha_{i,2}^{2e_{i,2,q}^a} - 1)/(\alpha_{i,2}^{2e_{i,2,q}^a} + 1)$ with $\alpha_{i,2} > 1$ and $e_{i,2,q}^a$ being the q -element of $e_{i,2}^a$, $\hat{\varrho}_{i,2,q}^a(t)$ is an adaptive gain, and

$$\sigma_{i,2,q}^a(\epsilon_{i,2,q}^a) = \frac{\hat{\varrho}_{i,2,q}^a \epsilon_{i,2,q}^a}{\sqrt{|\hat{\varrho}_{i,2,q}^a \epsilon_{i,2,q}^a|^2 + \varpi_{i,2}^2(t)}}. \quad (19)$$

Part II: For the position loop ($i \in \mathbb{V}$), to achieve the formation control of the heterogeneous UAVs-UGVs system (7), the following controllers are proposed as

$$\nu_{i,q} = -c_{i,1} \epsilon_{i,1,q} - \hat{\varrho}_{i,1,q} \sigma_{i,1,q}(\epsilon_{i,1,q}), \quad i \in \mathbb{V}, \quad q \in \mathbf{N}_1^{n_i}, \quad (20)$$

$$\bar{u}_{i,q} = \sum_{k=1}^{n_i} [B_{p,i}^{a,N}]_{q,k} (-c_{i,2} \epsilon_{i,2,k} - \hat{\varrho}_{i,2,k} \sigma_{i,2,k}(\epsilon_{i,2,k})), \quad (21)$$

$$\dot{\varrho}_{i,k} = \gamma_{i,k} |\epsilon_{i,k}| - \bar{\gamma}_{i,k} \varpi_{i,k}(t) \hat{\varrho}_{i,k}, \quad \hat{\varrho}_{i,k}(0) \geq 0, \quad (22)$$

where $k \in \mathbf{N}_1^2$; $e_{i,2}(t) = \dot{p}_i(t) - \nu_i(t)$ for $i \in \mathbb{M}_a$, $e_{i,2}(t) = \dot{p}_i(t) - \nu_i(t)$ for $i \in \mathbb{M}_g$; and other parameters and variables are defined similarly to Part I.

Remark 4: The control strategy proposed in this paper only requires simple parameter conditions of the funnel function, without the requirement in [29], [30], [31], [32], [33] to know all initial error conditions. This is mainly achieved through the ingenious non-singular nonlinear transformation in (15), with the detailed proof shown in Part I of the proof of *Theorem 1*.

Remark 5: Unlike the non-smooth asymptotic results in [37], [38], [39], the proposed control strategy is smooth, without the use of sign functions or function switching, making it more suitable for attitude tracking of UAVs in heterogeneous UAVs-UGVs systems. Additionally, the scalar adaptive law used in

this paper has a lower complexity compared to the literature that relies on complex function approximation laws [24], [25], [26], [27].

B. Stability Analysis

Theorem 1: For the heterogeneous UAVs-UGVs system (7). Under Assumptions 1–4, the robust distributed formation control strategy (13)–(22) guarantees that:

- 1) all closed-loop signals are bounded;
- 2) the funnel performance (11) is guaranteed;
- 3) the formation error and position tracking error asymptotically converge to zero, i.e., $\lim_{t \rightarrow \infty} \|e_1(t)\| = 0$ and $\lim_{t \rightarrow \infty} \|E(t)\| = 0$.

Proof. Part I: It is demonstrated that under the proposed attitude controller in the i -th UAV, the attitude tracking error remains confined within the funnel boundary and asymptotically converges to zero. Mathematically, the proposed attitude controller (13)–(18) ensures: 1) the funnel performance $e_1^a(t) \in \Omega_\mu^a := \{(e_1^a(t), t) \in \mathbf{R}^{3N} \times [0, \infty) : \|e_1^a(t)\| < \mu^a(t)\}$; 2) asymptotic convergence of the error $e_{i,k}^a$ ($i \in \mathbb{V}$, $k \in \mathbb{N}_1^2$) to zero; and 3) boundedness of all signals in attitude system of the i -th UAV. Since the similarity between this proof and the subsequent proof for the position system, the proof is omitted to ensure the conciseness and clarity of paper.

Part II: For the position system of (7), we begin by establishing the existence and uniqueness of a maximal solution to the closed-loop system over the time interval $\mathcal{T} := [0, t_M)$, where $t_M \in \{\mathbf{R}_+, \infty\}$. Hereafter, we may disregard the independent variable of the function whenever it is evident. Recalling the definition of $\check{e}_{i,k,q}$, $i \in \mathbb{V}$, $k \in \mathbb{N}_1^2$, $q \in \mathbb{N}_1^{n_i}$, taking its derivative yields

$$\dot{\check{e}}_{i,k,q} = \mu_{i,k,q}^{-1} (\varphi_{i,k,q} \dot{e}_{i,k,q} - \dot{\mu}_{i,k,q} \check{e}_{i,k,q}), \quad (23)$$

where $\varphi_{i,k,q} = 4\alpha_{i,k,q}^{2e_{i,k,q}} / ((\alpha_{i,k,q}^{2e_{i,k,q}} + 1)^2)$. According to (20), (17), and the definition of $h_{i,1,q}(e_{i,1,q})$, it can be checked that

$$\begin{aligned} \dot{\check{e}}_{i,1,q} &= \mu_{i,1,q}^{-1} \left[\varphi_{i,1,q} (\check{d}_i + \check{a}_{i0}) \left(-c_{i,1,q} e_{i,1,q} - \hat{\varrho}_{i,1,q} \right. \right. \\ &\quad \times \sigma_{i,1,q}(\epsilon_{i,1,q}) + \frac{1}{2} \log_{\alpha_{i,2}} \left(\frac{1 + \check{e}_{i,2,q} \mu_{i,2,q}}{1 - \check{e}_{i,2,q} \mu_{i,2,q}} \right) \\ &\quad + \varphi_{i,1,q} \sum_{l \in \mathbb{N}_i} \check{a}_{il} (x_{i,1,q} - x_{l,1,q} - \zeta_{il,q}) - \varphi_{i,1,q} \check{a}_{i0} f_{0,q} \\ &\quad \left. \left. - \varphi_{i,1,q} \sum_{l \in \mathbb{N}_i} \check{a}_{il} \left(-c_{l,1,q} \epsilon_{l,1,q} - \hat{\varrho}_{l,1,q} \sigma_{l,1,q}(\epsilon_{l,1,q}) \right) \right. \right. \\ &\quad \left. \left. + \frac{1}{2} \log_{\alpha_{i,2}} \left(\frac{1 + \check{e}_{l,2,q} \mu_{l,2,q}}{1 - \check{e}_{l,2,q} \mu_{l,2,q}} \right) \right) - \dot{\mu}_{i,1,q} \check{e}_{i,1,q} \right] \\ &= \check{h}_{i,1,q}(t, \check{e}_{i,1,q}, \check{e}_{i,2,q}, \check{e}_{l,1,q}, \check{e}_{l,2,q}, \hat{\varrho}_{i,1,q}, \hat{\varrho}_{l,1,q}), \quad (24) \end{aligned}$$

where $\check{e}_{l,k,n_i} = 0$ if $n_l \leq n_i$, and

$$\dot{\check{e}}_{i,2,q} = \mu_{i,2,q}^{-1} \left[\varphi_{i,2,q} \left(-c_{i,2,q} \epsilon_{i,2,q} - \hat{\varrho}_{i,2,q} \sigma_{i,2,q}(\epsilon_{i,2,q}) \right. \right.$$

$$\begin{aligned} &\left. + f_{i,q} + \Delta_{i,q} - \dot{\nu}_{i,q} \right) - \dot{\mu}_{i,2,q} \check{e}_{i,2,q} \Big] \\ &= \check{h}_{i,2,q}(t, \check{e}_{i,1,q}, \check{e}_{l,2,q}, \hat{\varrho}_{i,1,q}, \hat{\varrho}_{i,2,q}), \quad (25) \end{aligned}$$

with $\dot{\nu}_{i,q} = \frac{\partial \nu_{i,q}}{\partial e_{i,1,q}} \dot{e}_{i,1,q} + \frac{\partial \nu_{i,q}}{\partial \mu_{i,1,q}} \dot{\mu}_{i,1,q} + \frac{\partial \nu_{i,q}}{\partial \hat{\varrho}_{i,1,q}} \dot{\hat{\varrho}}_{i,1,q} + \frac{\partial \nu_{i,q}}{\partial \varpi_{i,1}} \dot{\varpi}_{i,q}$, where $f_{i,q}$ and $\Delta_{i,q}$ are the q -th element of $\mathbf{f}_i$ and Δ_i , where $\mathbf{f}_i = -g\mathbf{o}_3 \in \mathbf{R}^3$ and $\Delta_i = \Delta_{p,i}^a \in \mathbf{R}^3$ if $i \in \mathbb{M}_a$; $\mathbf{f}_i = \mathbf{f}_i^g \in \mathbf{R}^2$ and $\Delta_i = \Delta_i^g \in \mathbf{R}^2$, if $i \in \mathbb{M}_g$. Collecting $\check{\mathbf{e}} = [\check{\mathbf{e}}_1^T, \check{\mathbf{e}}_2^T]^T$ with $\check{\mathbf{e}}_k = [\check{e}_{1,k}^T, \hat{\varrho}_{1,k}^T, \dots, \check{e}_{N,k}^T, \hat{\varrho}_{N,k}^T]^T$, for $k \in \mathbb{N}_1^2$, where $\check{e}_{i,k} = [\check{e}_{i,k,1}, \dots, \check{e}_{i,k,n_i}]^T$ and $\hat{\varrho}_{i,k} = [\hat{\varrho}_{i,k,1}, \dots, \hat{\varrho}_{i,k,n_i}]^T$, then (24)–(25) may be compactly formulated as

$$\dot{\check{\mathbf{e}}} = \check{\mathbf{h}}(t, \check{\mathbf{e}}) = \begin{bmatrix} \check{\mathbf{h}}_1(t, \check{\mathbf{e}}_1, \check{\mathbf{e}}_2) \\ \check{\mathbf{h}}_2(t, \check{\mathbf{e}}_1, \check{\mathbf{e}}_2) \end{bmatrix}. \quad (26)$$

Define $\bar{\Omega}_\mu = \{(\check{\mathbf{e}}, t) \in \mathbb{R}^{2N\bar{n}} \times [0, \infty) : |\check{e}_{i,k,q}| < 1, i \in \mathbb{V}, k \in \mathbb{N}_1^2, q \in \mathbb{N}_1^{n_i}\}$, which is nonempty and open. It holds from Section III-A that $\mu_{i,k,q}(0) = 1$. From the definition of $h_{i,k,q}(\cdot)$ and $\check{e}_{i,k,q}$, we infer that $h_{i,k,q}(e_{i,k,q}(0)) < \mu_{i,k,q}(0) = 1$ for any $e_{i,k,q}(0)$, which gives $\check{e}_{i,k,q}(0) \in (-1, 1)$, thus, $\check{\mathbf{e}}(t)$ initially belongs to $\bar{\Omega}_\mu$, i.e., $\check{\mathbf{e}}(0) \in \bar{\Omega}_\mu$. Secondly, for $i \in \mathbb{M}_a$, according to the statement described in Part I, the attitude can track the reference attitude with the preset funnel performance, and all signals of the attitude system are bounded. Recalling that $\tilde{\mathbf{u}}_{p,i}^a = \mathbf{u}_{z,i}^a(\mathbb{R}^{\text{EB}}(\boldsymbol{\eta}_i^a) - \mathbb{R}^{\text{EB}}(\boldsymbol{\eta}_{r,i}^a))\mathbf{o}_3 + \mathbf{u}_{z,i}^a(B_{p,i}^{a,N})^{-1}B_{p,i}^{a,\Delta}\mathbb{R}^{\text{EB}}(\boldsymbol{\eta}_i^a)\mathbf{o}_3$, it follows from Part I and Assumption 3 that $\Delta_i^g \in \mathcal{L}_\infty$. Therefore, Δ_i is piecewise continuous and bounded. Further, in relation to (26) under Assumptions 2–4, the piecewise continuity and boundedness of $\mu_{i,k,q}$, $\dot{\mu}_{i,k,q}$, Δ_i , $\check{a}_{il}$, $\check{a}_{il}$, $\varpi_{i,q}$, and $\dot{\varpi}_{i,q}$ can be ensured over $\mathcal{T}$, along with the smooth functions $\mathbf{f}_i$ ($i \in \mathbb{V}$), one can be checked that $\check{\mathbf{h}}$ is locally Lipschitz on $\check{\mathbf{e}}$. Hence, the existence and uniqueness of a maximal solution $\check{\mathbf{e}} : \mathcal{T} \rightarrow \bar{\Omega}_\mu$ is guaranteed by applying the Theorem 1 in [22].

Subsequently, we show that $t_M = \infty$. Let's now define the Lyapunov function candidate

$$V_1 = \sum_{i=1}^N \sum_{q=1}^{n_i} \left(\frac{\epsilon_{i,1,q}^2}{2(\check{d}_i + \check{a}_{i0})} + \frac{\check{\varrho}_{i,1,q}^2}{2\gamma_{i,1}\varsigma_{i,1,q}} \right), \quad (27)$$

where $\check{\varrho}_{i,1,q} = \varsigma_{i,1,q} \hat{\varrho}_{i,1,q} - \varrho_{i,1,q}$ with constants $\varrho_{i,1,q}, \varsigma_{i,1,q} > 0$ defined later. Based on the definitions of $\epsilon_{i,1,q}(\cdot)$ and $h_{i,1,q}(\cdot)$, computing the derivative of V_1 yields

$$\begin{aligned} \dot{V}_1 &= \sum_{i=1}^N \sum_{q=1}^{n_i} \left[\frac{\epsilon_{i,1,q} r_{i,1,q} \mu_{i,1,q}^{-1}}{\check{d}_i + \check{a}_{i0}} (\varphi_{i,1,q} \dot{e}_{i,1,q} - \dot{\mu}_{i,1,q} \check{e}_{i,1,q}) \right. \\ &\quad \left. + \frac{\check{\varrho}_{i,1,q}}{\gamma_{i,1}} \dot{\hat{\varrho}}_{i,1,q} \right], \quad (28) \end{aligned}$$

where $r_{i,1,q} = 2/(1 - \check{e}_{i,1,q}^2)$. According to (9), it yields that $\dot{e}_{i,1,q} = (\check{d}_i + \check{a}_{i0})\dot{x}_{i,1,q} - \sum_{l \in \mathbb{N}_i} \check{a}_{il}\dot{x}_{l,1,q} + \sum_{l \in \mathbb{N}_i} \check{a}_{il}(x_{i,1,q} - x_{l,1,q} - \zeta_{il,q})$, which gives

$$\dot{V}_1 = \sum_{i=1}^N \sum_{q=1}^{n_i} \left[\epsilon_{i,1,q} r_{i,1,q} \mu_{i,1,q}^{-1} \varphi_{i,1,q} \nu_{i,q} + \frac{1}{2} \epsilon_{i,1,q} r_{i,1,q} \mu_{i,1,q}^{-1} \right.$$

$$\begin{aligned} & \times \varphi_{i,1,q} \log_{\alpha_{i,2}} \left(\frac{1 + \check{e}_{i,2,q} \mu_{i,2,q}}{1 - \check{e}_{i,2,q} \mu_{i,2,q}} \right) - \frac{\epsilon_{i,1,q} r_{i,1,q} \mu_{i,1,q}^{-1}}{\check{d}_i + \check{a}_{i0}} \\ & \times \left(\sum_{l \in \bar{\mathbb{N}}_i} \check{a}_{il} \dot{x}_{l,1,q} - \sum_{l \in \bar{\mathbb{N}}_i} \dot{\check{a}}_{il} (x_{i,1,q} - x_{l,1,q} - \zeta_{il,q}) \right. \\ & \left. + \dot{\mu}_{i,1,q} \check{e}_{i,1,q} \right) + \frac{\tilde{\varrho}_{i,1,q}}{\gamma_{i,1}} \dot{\hat{\varrho}}_{i,1,q} \Bigg], \quad (29) \end{aligned}$$

where $\dot{x}_{l,1,q} = \frac{1}{2} \log_{\alpha_{i,2}}((1 + \check{e}_{l,2,q} \mu_{l,2,q}) / (1 - \check{e}_{l,2,q} \mu_{l,2,q})) - c_{l,1,q} \epsilon_{l,1,q} - \hat{\varrho}_{l,1,q} \sigma_{l,1,q}(\epsilon_{l,1,q})$ according to (15), (20), and the definition of $e_{i,2}$, $\dot{x}_{0,1,q} = f_{0,q}$. Note that the piecewise continuity and boundedness of f_0 , $\mu_{i,1,q}$, $\dot{\mu}_{i,1,q}$, $\check{a}_{il}$, $\dot{\check{a}}_{il}$, $\varpi_{i,q}$, $\dot{\varpi}_{i,q}$, and $\varphi_{i,1,q}$, together with the unique maximal solution $\tilde{e} : \mathcal{T} \rightarrow \bar{\Omega}_\mu$, it follows from the *extreme value theorem* that

$$\begin{aligned} & \left| \frac{1}{2} r_{i,1,q} \mu_{i,1,q}^{-1} \varphi_{i,1,q} \log_{\alpha_{i,2}} \left(\frac{1 + \check{e}_{i,2,q} \mu_{i,2,q}}{1 - \check{e}_{i,2,q} \mu_{i,2,q}} \right) - \frac{r_{i,1,q} \mu_{i,1,q}^{-1}}{\check{d}_i + \check{a}_{i0}} \right. \\ & \times \left(\sum_{l \in \bar{\mathbb{N}}_i} \check{a}_{il} \dot{x}_{l,1,q} - \sum_{l \in \bar{\mathbb{N}}_i} \dot{\check{a}}_{il} (x_{i,1,q} - x_{l,1,q} - \zeta_{il,q}) \right. \\ & \left. \left. + \dot{\mu}_{i,1,q} \check{e}_{i,1,q} \right) \right| \leq \varrho_{i,1,q}, \quad (30) \end{aligned}$$

for an unknown positive scalar $\varrho_{i,1,q}$, $\forall t \in \mathcal{T}$. Then, $\dot{V}_1$ becomes

$$\begin{aligned} \dot{V}_1 \leq & \sum_{i=1}^N \sum_{q=1}^{n_i} \left(\epsilon_{i,1,q} r_{i,1,q} \mu_{i,1,q}^{-1} \varphi_{i,1,q} \nu_{i,q} + |\epsilon_{i,1,q}| \varrho_{i,1,q} \right. \\ & \left. + \frac{\tilde{\varrho}_{i,1,q}}{\gamma_{i,1}} \dot{\hat{\varrho}}_{i,1,q} \right), \quad \forall t \in \mathcal{T}. \quad (31) \end{aligned}$$

Taking (20) into (31) yields

$$\begin{aligned} \dot{V}_1 \leq & \sum_{i=1}^N \sum_{q=1}^{n_i} \left(-c_{i,1} r_{i,1,q} \mu_{i,1,q}^{-1} \varphi_{i,1,q} |\epsilon_{i,1,q}|^2 + |\epsilon_{i,1,q}| \varrho_{i,1,q} \right. \\ & \left. - \frac{r_{i,1,q} \mu_{i,1,q}^{-1} \varphi_{i,1,q} |\hat{\varrho}_{i,1,q} \epsilon_{i,1,q}|^2}{\sqrt{|\hat{\varrho}_{i,1,q} \epsilon_{i,1,q}|^2 + \varpi_{i,1}^2}} + \frac{\tilde{\varrho}_{i,1,q}}{\gamma_{i,1}} \dot{\hat{\varrho}}_{i,1,q} \right), \quad (32) \end{aligned}$$

$\forall t \in \mathcal{T}$. It follows from the existence of a unique maximal solution $\tilde{e} : \mathcal{T} \rightarrow \bar{\Omega}_\mu$ that $\varphi_{i,1,q} > 0$ and $|(1 - \check{e}_{i,1,q}^2)^{-1}| \geq 1$, which indicates that $r_{i,1,q} \geq 2 > 0$, $\forall t \in \mathcal{T}$. According to the definition of $\varphi_{i,1,q}$, there is a constant $\varphi_{i,1,q} > 0$ ensuring $\varphi_{i,1,q}(t) \geq \varphi_{i,1,q}$, $\forall t \in \mathcal{T}$. Therefore, one gets $r_{i,1,q} \mu_{i,1,q}^{-1} \varphi_{i,1,q} \geq \varsigma_{i,1,q}$ with a constant $\varsigma_{i,1,q} > 0$, $\forall t \in \mathcal{T}$. According to (22), it yields $\hat{\varrho}_{i,1,q} \geq 0$. Then, based on $0 \leq |z| - \frac{|z|^2}{\sqrt{|z|^2 + \varpi^2}} \leq \varpi$ [42], for all $z \in \mathbf{R}$ and $\varpi \in \mathbf{R}_+$ and $\tilde{\varrho}_{i,1,q} = \varsigma_{i,1,q} \hat{\varrho}_{i,1,q} - \varrho_{i,1,q}$, one obtains

$$\begin{aligned} \dot{V}_1 \leq & \sum_{i=1}^N \sum_{q=1}^{n_i} \left(-c_{i,1} \varsigma_{i,1,q} |\epsilon_{i,1,q}|^2 - |\epsilon_{i,1,q}| \tilde{\varrho}_{i,1,q} \right. \\ & \left. + \varsigma_{i,1,q} \varpi_{i,1} + \frac{\tilde{\varrho}_{i,1,q}}{\gamma_{i,1}} \dot{\hat{\varrho}}_{i,1,q} \right), \quad \forall t \in \mathcal{T}. \quad (33) \end{aligned}$$

Incorporating (22) into (33) gets

$$\begin{aligned} \dot{V}_1 \leq & \sum_{i=1}^N \sum_{q=1}^{n_i} \left[-c_{i,1} \varsigma_{i,1,q} |\epsilon_{i,1,q}|^2 + \varsigma_{i,1,q} \varpi_{i,1} \right. \\ & \left. - \frac{\tilde{\gamma}_{i,1} \varpi_{i,1}}{\gamma_{i,1}} \tilde{\varrho}_{i,1,q} \hat{\varrho}_{i,1,q} \right], \quad \forall t \in \mathcal{T}. \quad (34) \end{aligned}$$

From the definition of $\tilde{\varrho}_{i,1,q}$ and the Young's inequality, one gets

$$-\frac{\tilde{\gamma}_{i,1} \varpi_{i,1}}{\gamma_{i,1}} \tilde{\varrho}_{i,1,q} \hat{\varrho}_{i,1,q} \leq \frac{\tilde{\gamma}_{i,1} \varpi_{i,1}}{4\gamma_{i,1} \varsigma_{i,1,q}} \varrho_{i,1,q}^2, \quad (35)$$

it thus derives that

$$\dot{V}_1 \leq \sum_{i=1}^N \sum_{q=1}^{n_i} \left[-c_{i,1} \varsigma_{i,1,q} |\epsilon_{i,1,q}|^2 + \pi_{i,1,q} \varpi_{i,1} \right], \quad (36)$$

$\forall t \in \mathcal{T}$, where $\pi_{i,1,q} = \varsigma_{i,1,q} + \tilde{\gamma}_{i,1} \varrho_{i,1,q}^2 / (4\gamma_{i,1} \varsigma_{i,1,q})$ is a positive constant. Then, it deduces that $\dot{V}_1 < 0$ when $|\epsilon_{i,1,q}| > \sqrt{\pi_{i,1,q} \varpi_{i,1} / (c_{i,1} \varsigma_{i,1,q})}$, where $\varpi_{i,1,q} = \sup_{t \in \mathcal{T}} \{\varpi_{i,1,q}(t)\} > 0$. Hence,

$$|\epsilon_{i,1,q}(t)| \leq \bar{\epsilon}_{i,1,q} := \max \left\{ |\epsilon_{i,1,q}(0)|, \sqrt{\frac{\pi_{i,1,q} \varpi_{i,1}}{c_{i,1} \varsigma_{i,1,q}}} \right\}, \quad (37)$$

$\forall t \in \mathcal{T}$. By virtue of the definition of $\epsilon_{i,1,q}$ and subsequently using the inverse logarithmic function, one obtains

$$\check{e}_{i,1,q} \in \Omega_{i,1,q} := \left[\frac{e^{-\bar{\epsilon}_{i,1,q}} - 1}{e^{-\bar{\epsilon}_{i,1,q}} + 1}, \frac{e^{\bar{\epsilon}_{i,1,q}} - 1}{e^{\bar{\epsilon}_{i,1,q}} + 1} \right] \subset (-1, 1), \quad (38)$$

$\forall t \in \mathcal{T}$. Then, one can infer that $e_{i,1,q} \in \mathcal{L}_\infty$ and $\tilde{\varrho}_{i,1,q} \in \mathcal{L}_\infty$ based on (27) and (36), $\forall t \in \mathcal{T}$. Due to $\varrho_{i,1,q}$ is a finite constant and $\tilde{\varrho}_{i,1,q} = \varsigma_{i,1,q} \hat{\varrho}_{i,1,q} - \varrho_{i,1,q}$, one can be checked that $\hat{\varrho}_{i,1,q}(t) \in \mathcal{L}_\infty$, $\forall t \in \mathcal{T}$. Hence, $\check{e}_{i,1,q} := [\check{e}_{i,1,q}, \hat{\varrho}_{i,1,q}]^T \in \bar{\Omega}_{i,1,q} = \{(\check{e}_{i,1,q}(t), t) \in \mathbf{R}^2 \times [0, \infty) : \check{e}_{i,1,q} \in \Omega_{i,1,q}\}$ holds for all $t \in \mathcal{T}$. And we obtain that $\nu_i(t) \in \mathcal{L}_\infty$, $\forall t \in \mathcal{T}$.

Next, define the Lyapunov function candidate as

$$V_2 = \sum_{i=1}^N \sum_{q=1}^{n_i} \left(\frac{1}{2} \epsilon_{i,2,q}^2 + \frac{\tilde{\varrho}_{i,2,q}^2}{2\gamma_{i,2} \varsigma_{i,2,q}} \right), \quad (39)$$

where $\tilde{\varrho}_{i,2,q} = \varsigma_{i,2,q} \hat{\varrho}_{i,2,q} - \varrho_{i,2,q}$ with $\varsigma_{i,2,q} > 0$ being the lower bounded of $r_{i,2,q} \mu_{i,2,q}^{-1} \varphi_{i,2,q}$, $\varrho_{i,2,q} > 0$ is a constant and $r_{i,2,q} = 2/(1 - \check{e}_{i,2,q}^2)$. By following the lines from (28) to (36), $\dot{V}_2$ can be directly bounded as

$$\dot{V}_2 \leq \sum_{i=1}^N \sum_{q=1}^{n_i} \left[-c_{i,2} \varsigma_{i,2,q} |\epsilon_{i,2,q}|^2 + \pi_{i,2,q} \varpi_{i,2,q} \right], \quad \forall t \in \mathcal{T}, \quad (40)$$

where $\pi_{i,2,q} = \varsigma_{i,2,q} + \tilde{\gamma}_{i,2} \varrho_{i,2,q}^2 / (4\gamma_{i,2} \varsigma_{i,2,q})$. Then, by defining $\varpi_{i,2,q} = \sup_{t \in \mathcal{T}} \{\varpi_{i,2,q}(t)\} > 0$, we have

$$|\epsilon_{i,2,q}(t)| \leq \bar{\epsilon}_{i,2,q} := \max \left\{ |\epsilon_{i,2,q}(0)|, \sqrt{\frac{\pi_{i,2,q} \varpi_{i,2,q}}{c_{i,2} \varsigma_{i,2,q}}} \right\}, \quad (41)$$

$$\check{e}_{i,2,q} \in \Omega_{i,2,q} := \left[\frac{e^{-\bar{\epsilon}_{i,2,q}} - 1}{e^{-\bar{\epsilon}_{i,2,q}} + 1}, \frac{e^{\bar{\epsilon}_{i,2,q}} - 1}{e^{\bar{\epsilon}_{i,2,q}} + 1} \right] \subset (-1, 1), \quad (42)$$

$\forall t \in \mathcal{T}$, can be obtained. Similarly, one can also deduce that $e_{i,2,q}, \hat{e}_{i,2,q} \in \mathcal{L}_\infty, \forall t \in \mathcal{T}$. Hence, $\bar{e}_{i,2,q} := [\bar{e}_{i,2,q}, \hat{e}_{i,2,q}]^T \in \bar{\Omega}_{i,2,q} = \{(\bar{e}_{i,2,q}(t), t) \in \mathbf{R}^2 \times [0, \infty) : \bar{e}_{i,2,q} \in \Omega_{i,2,q}\}$ holds for all $t \in \mathcal{T}$. Thus, it can be checked that $\mathbf{u}_i(t) \in \mathcal{L}_\infty, \forall t \in \mathcal{T}$. The next procedure is the extension of t_M to ∞ . Defining $\bar{\Omega}'_1 = \bar{\Omega}_{1,1,1} \times \cdots \times \bar{\Omega}_{1,1,n_1} \times \cdots \times \bar{\Omega}_{N,1,1} \times \cdots \times \bar{\Omega}_{N,1,n_N} \times \bar{\Omega}_{1,2,1} \times \cdots \times \bar{\Omega}_{1,2,n_1} \times \cdots \times \bar{\Omega}_{N,2,1} \times \cdots \times \bar{\Omega}_{N,2,n_N}$, it follows that $\bar{e}(t) \in \bar{\Omega}'_1 \subset \bar{\Omega}_\mu$ for all $t \in \mathcal{T}$. Then, let's assume $t_M < \infty$, based on the Proposition 1 in [22], there exists a time instant $t_1 \in \mathcal{T}$ such that $\bar{e}(t_1) \notin \bar{\Omega}'_1$, which results in a contradiction. Hence, we arrive that $t_M = \infty$ and all the closed-loop signals are bounded.

Hence, we have $-1 < \bar{e}_{i,k,q}(t) < 1, \forall t \geq 0$ and $\nu_i(t)$ and $\mathbf{u}_i(t)$ are uniformly bounded, for $i \in \mathbb{V}, k \in \mathbb{N}_1^2$, and $q \in \mathbb{N}_1^{n_i}$. From the definition of $\bar{e}_{i,k,q}(t)$ and the properties of $\mu_{i,k,q}(t)$, it is deduced that $h_{i,k,q}(e_{i,k,q})$ converges to $\{h_{i,k,q}(e_{i,k,q}) \in \mathbf{R} : |h_{i,k,q}(e_{i,k,q})| < \mu_{i,k,q}(t)\}, \forall t \geq 0$, and converges to $\{h_{i,k,q}(e_{i,k,q}) \in \mathbf{R} : |h_{i,k,q}(e_{i,k,q})| < \mu_{i,k,q}^\infty\}$ as $t \rightarrow \infty$. Based on the definition of $h_{i,k,q}(\cdot)$, one obtains

$$|e_{i,k,q}| < \mu'_{i,k,q}(t) := \frac{1}{2} \log_{\alpha_{i,k}} \left(\frac{1 + \mu_{i,k,q}(t)}{1 - \mu_{i,k,q}(t)} \right), \quad (43)$$

$\forall t \geq 0$, which further yields

$$\|e_1(t)\| < \mu(t) := \sqrt{3N} \bar{\mu}(t), \quad \forall t \geq 0, \quad (44)$$

with $\bar{\mu}(t) = \max_{i \in \mathbb{V}, q \in \mathbb{N}_1^{n_i}} \mu'_{i,1,q}(t)$. Hence, the funnel performance $e_1(t) \in \Omega_\mu$ is guaranteed.

Then, we define $V = V_1 + V_2$, it follows from (36) and (40) that $\dot{V} \leq \sum_{i=1}^N \sum_{q=1}^{n_i} [-c_{i,1} \varsigma_{i,1,q} |\epsilon_{i,1,q}|^2 - c_{i,2} \varsigma_{i,2,q} |\epsilon_{i,2,q}|^2 + \pi_{i,1,q} \varpi_{i,1} + \pi_{i,2,q} \varpi_{i,2}]$, integrating it from 0 to t obtains

$$\begin{aligned} V(t) &\leq V(0) - \sum_{k=1}^2 \sum_{i=1}^N \sum_{q=1}^{n_i} \int_0^t c_{i,k} \varsigma_{i,k,q} \epsilon_{i,k,q}^2(\iota) d\iota \\ &\quad + \sum_{k=1}^2 \sum_{i=1}^N \sum_{q=1}^{n_i} \int_0^t \pi_{i,k,q} \varpi_{i,k}(\iota) d\iota \\ &\leq V(0) + \delta, \end{aligned} \quad (45)$$

with $\delta = \sum_{k=1}^2 \sum_{i=1}^N \sum_{q=1}^{n_i} \pi_{i,k,q} \varpi_{i,k}^* > 0$ being a constant. Recalling the definition of V , it follows that $\epsilon_{i,k,q} \in \mathcal{L}_\infty \cap \mathcal{L}_2$, for $i \in \mathbb{V}, k \in \mathbb{N}_1^2$, and $q \in \mathbb{N}_1^{n_i}$. Since $\dot{\epsilon}_{i,k,q} = r_{i,k,q} \mu_{i,k,q}^{-1}(\varphi_{i,k,q} \dot{e}_{i,k,q} - \dot{\mu}_{i,k,q} \bar{e}_{i,k,q})$, it can be checked that $\dot{\epsilon}_{i,k,q} \in \mathcal{L}_\infty$. By utilizing the Barbalat's lemma, one obtains that $\lim_{t \rightarrow \infty} \epsilon_{i,k,q}(\bar{e}_{i,k,q}) = 0$. Hence, noting that $\epsilon_{i,k,q}(0) = 0$ and $h_{i,k,q}(0) = 0$, $\lim_{t \rightarrow \infty} e_{i,k,q}(t) = 0$ is achieved. Further, since $\mathbb{H}(t)$ is non-singular when Assumptions 1 and 2 are met, it follows from (10) that

$$\|\mathbf{E}(t)\| \leq \frac{\|e_1(t)\|}{\text{svd}(\mathbb{H}(t))}, \quad \forall t \geq 0, \quad (46)$$

where $\text{svd}(\cdot)$ stands for the minimum singular value. Since $\lim_{t \rightarrow \infty} e_{i,k,q}(t) = 0$, which implies $\lim_{t \rightarrow \infty} \|e_1(t)\| = 0$, it is checked from (46) that $\lim_{t \rightarrow \infty} \|\mathbf{E}(t)\| = 0$. A monotonic relationship between communication noise intensity and transient position tracking error magnitude is established in (46). Notably,

the proposed distributed control strategy guarantees asymptotic convergence under the communication noises, all position tracking errors decay to zero despite finite-time error amplification. This property formally decouples steady-state formation accuracy from communication noises, thereby asymptotically rejecting their destabilizing effects.

Part III: For $i \in \mathbb{M}_g$, the control torque τ_i^g also influences the attitude of the UGV. Therefore, it is necessary to analyze the stability of the UGV's attitude. First, the kinematic relationship between the sensor position and the i -th UGV is as follows:

$$\dot{\mathbf{p}}_i^g = \begin{bmatrix} \cos \eta_i^g & -d_i \sin \eta_i^g \\ \sin \eta_i^g & d_i \cos \eta_i^g \end{bmatrix} \begin{bmatrix} v_i^g \\ \omega_i^g \end{bmatrix}. \quad (47)$$

According to $\det([\cos \eta_i^g, -d_i \sin \eta_i^g; \sin \eta_i^g, d_i \cos \eta_i^g]) \neq 0$ and $\mathbf{E} = \mathbf{p} - \mathbf{1}_N \otimes \mathbf{p}_0 - \boldsymbol{\zeta}$, one obtains

$$\begin{bmatrix} v_i^g \\ \omega_i^g \end{bmatrix} = \begin{bmatrix} \cos \eta_i^g & \sin \eta_i^g \\ -\frac{1}{d_i} \sin \eta_i^g & \frac{1}{d_i} \cos \eta_i^g \end{bmatrix} (\dot{\mathbf{p}}_0 - \dot{\mathbf{E}}_i). \quad (48)$$

It yields

$$\omega_i^g = \frac{1}{d_i} \left(-\sin \eta_i^g (\dot{p}_{0,x} - \dot{E}_{i,x}) + \cos \eta_i^g (\dot{p}_{0,y} - \dot{E}_{i,y}) \right).$$

Define the heading angle tracking error as $e_{\eta,i} = \eta_0 - \eta_i^g$, where $\eta_0 = \arctan2(\dot{p}_{0,y}/\dot{p}_{0,x})$ is the reference heading angle. Its dynamics is as follows:

$$\dot{e}_{\eta,i} = \dot{\eta}_0 - \frac{1}{d_i} \left(-\sin \eta_i^g (\dot{p}_{0,x} - \dot{E}_{i,x}) + \cos \eta_i^g (\dot{p}_{0,y} - \dot{E}_{i,y}) \right).$$

Using $\dot{\mathbf{p}}_0 = v_0 [\cos \eta_0, \sin \eta_0]^T$ with $v_0 = \|\dot{\mathbf{p}}_0\|$, we have

$$\dot{e}_{\eta,i} = \dot{\eta}_0 - \frac{v_0}{d_i} \sin e_{\eta,i} + \frac{1}{d_i} \left(\dot{E}_{i,x} \sin \eta_i^g - \dot{E}_{i,y} \cos \eta_i^g \right).$$

That is, $\dot{e}_{\eta,i} = \dot{\eta}_0 - (v_0/d_i) \sin e_{\eta,i} + \delta_i(t)$, where $\delta_i(t) = (\dot{E}_{i,x} \sin \eta_i^g - \dot{E}_{i,y} \cos \eta_i^g)/d_i$ is the coupling term caused by the position tracking error. It follows from Part II that $e_1(t) \in \Omega_\mu$ and $\lim_{t \rightarrow \infty} \|\mathbf{E}(t)\| = 0$, then $\mathbf{E}_i$ and its derivatives are bounded. Since $\dot{E}_{i,x}, \dot{E}_{i,y} \in \mathcal{L}_\infty$ and $\dot{\eta}_0 \in \mathcal{L}_\infty$ from Assumption 4, then $\delta_i(t) \in \mathcal{L}_\infty$. For the attitude subsystem, choose the Lyapunov function candidate as $V_{\eta,i} = 1 - \cos e_{\eta,i} \geq 0$, where $V_{\eta,i} = 0$ if and only if $e_{\eta,i} = 0$, whose derivative is

$$\begin{aligned} \dot{V}_{\eta,i} &= -\frac{v_0}{d_i} (\sin e_{\eta,i})^2 + \sin e_{\eta,i} (\dot{\eta}_0 + \delta_i(t)) \\ &\leq -\frac{v_0}{d_i} |\sin e_{\eta,i}|^2 + |\sin e_{\eta,i}| (|\dot{\eta}_0| + |\delta_i(t)|). \end{aligned} \quad (49)$$

It follows from Part II that $\dot{\eta}_0, \delta_i(t) \in \mathcal{L}_\infty$, by invoking Young's inequality and following a procedure analogous to that in (36)–(37), we infer that for each $i \in \mathbb{M}_g$, the attitude subsystem is uniform boundedness.

Remark 6: By modifying the estimation error and leveraging the properties of the adaptive law along with some appropriate scaling techniques, the designed controller incorporates fewer nonlinear components. More specifically, the work [38] proposed the following asymptotic consensus tracking funnel control scheme: $\nu_{i,q} = -c_{i,1} r_{i,1,q} \epsilon_{i,1,q} - \mu_{i,1,q} \hat{e}_{i,1,q} \text{sign}(r_{i,1,q} \epsilon_{i,1,q})$ and $\hat{e}_{i,1,q} = \gamma_{i,1} |r_{i,1,q} \epsilon_{i,1,q}|$, but the

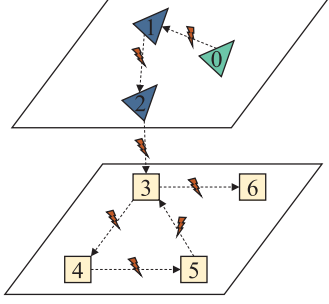


Fig. 4. Interaction topology of the heterogeneous UAVs-UGVs system: the triangle represents the UAV, the rectangle denotes the UGV, and the 0 represents the leader.

complex nonlinear functions $r_{i,1,q} = 2/(1 - \tilde{e}_{i,1,q}^2)$ and $\mu_{i,1,q}$, as well as the switching function $\text{sign}(r_{i,1,q}\epsilon_{i,1,q})$, do not exist in the proposed controller in (20), resulting in lower structural complexity and smooth nature, which makes it easier to implement in vehicles.

Remark 7: The sensitivity of the proposed strategy with respect to the parameters is herein demonstrated. The formation error trajectory is regulated by the adjustments of the funnel boundary via parameters $\alpha_{i,k}$ and $\mu_{i,k,q}$. $\alpha_{i,k}$ plays a crucial role in determining the transient performance by expediting the convergence towards the steady-state boundaries. While remaining parameters are freely design, their selection imposes critical performance trade-offs: Larger $c_{i,k}$ and $\gamma_{i,k}$ values enhance convergence rates at the expense of transient control effort magnitude, whereas reduced parameters diminish control effort magnitude while slowing error attenuation.

IV. SIMULATION STUDY

This section utilizes numerical simulations with a group of six vehicles (including two UAVs and four UGVs) to validate the effectiveness of the proposed strategy. The interaction topology satisfying Assumption 1 is depicted in Fig. 4.

A. Simulation Conditions and Results

The trajectory of the leader is generated by $\dot{p}_0 = [1.2, 0.4 \cos(0.4t), -0.2p_{0,z} + 2.5]^T$ satisfying Assumption 4 with $p_0(0) = [0, 0, 2]^T$ m, $\psi_{r,i} = 0$ rad, and the vehicles are required to keep a formation configuration $\mathbb{F}_t$ as: $\zeta_{10} = [0, 2, 0]^T$, $\zeta_{21} = [0, -4, 0]^T$, $\zeta_{32} = [0, 4, 0]^T$, $\zeta_{35} = [0, 4, 0]^T$, $\zeta_{43} = [-2, -2, 0]^T$, $\zeta_{54} = [2, -2, 0]^T$, and $\zeta_{63} = [2, -2, 0]^T$. According to [4], the physical parameters of the heterogeneous UAVs-UGVs system are set as $m_i^a = 2.5$ kg, $g = 9.81$ m/s², $B_{p,i}^{a,N} = 0.4\mathbf{I}_3$, $\mathcal{I}_i^{a,N} = \text{diag}(10.0, 8.2, 14.8)10^{-3}$ kg·m², $B_{\eta,i}^{a,N} = \text{diag}(100, 122, 68)$, $i \in \mathbb{M}_a := \{1, 2\}$, $m_i^g = 23$ kg, $\mathcal{I}_i^{g,N} = 1$ kg·m², $r_i = 0.02$ m, $s_i = 0.2$ m, and $d_i = 0.15$ m, $i \in \mathbb{M}_g := \{3, 4, 5, 6\}$. The robust controller parameters are $c_{i,1} = 1$, $c_{i,2} = 25$, $\gamma_{i,1} = 1$, $\gamma_{i,2} = 40$, $\bar{\gamma}_{i,1} = \bar{\gamma}_{i,2} = 0.1$, $\alpha_{i,1} = \alpha_{i,2} = 1.2$, $\varpi_{i,1} = \varpi_{i,2} = \exp(-0.01t) \in \mathcal{L}_1$, $\alpha_{i,1}^a = \alpha_{i,2}^a = 1.5$, $c_{1,1}^a = 10$, $c_{1,2}^a = 30$, $c_{2,1}^a = 1$, $c_{2,2}^a = 25$, $\gamma_{i,1}^a = 2$, $\gamma_{i,2}^a = 40$, $i \in \mathbb{M}_a$, $c_{i,1} = 5$, $c_{i,2} = 25$, $\gamma_{i,1} = 1$, $\gamma_{i,2} = 20$,

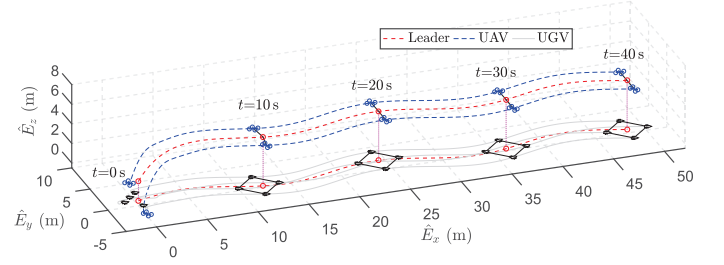


Fig. 5. 3D trajectories of the heterogeneous UAVs-UGVs system.

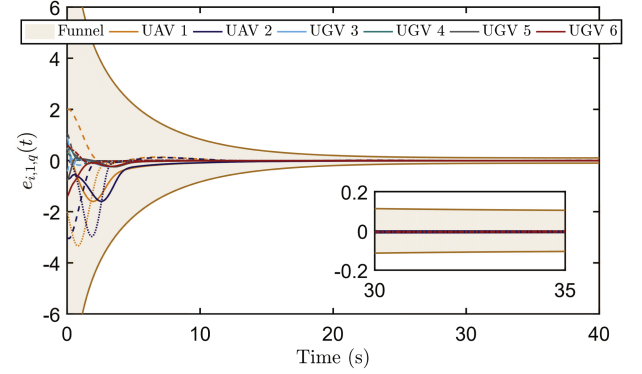


Fig. 6. Trajectories of formation errors [m]: the solid line is the $\hat{E}_x$ -axis component, the dashed line is the $\hat{E}_y$ -axis component, and the dotted line represents the $\hat{E}_z$ -axis component.

$i \in \mathbb{M}_g$. The funnel functions are chosen as $\mu_{i,1,q} = (1 - o_{i,1,q})e^{-0.2t} + o_{i,1,q}$, $\mu_{i,2,q} = (1 - o_{i,2,q})e^{-0.5t} + o_{i,2,q}$, $\mu_{i,1,q}^a = (1 - o_{i,1,q}^a)e^{-2t} + o_{i,1,q}^a$, and $\mu_{i,2,q}^a = \mu_{i,2,q}$ with $o_{i,1,q} = h_{i,1,q}(0.1)$, $o_{i,2,q} = h_{i,2,q}(0.5)$, and $o_{i,1,q}^a = h_{i,1,q}^a(0.1)$ such that $\mu_{i,1,q}(0) = \mu_{i,2,q}(0) = \mu_{i,1,q}^a(0) = \mu_{i,2,q}^a(0) = 1$. The initial values are given as follows: $p_1^a(0) = [0.5, 4, 0]^T$ m, $p_2^a(0) = [-0.2, -3, 0]^T$ m, $\eta_1^a(0) = \eta_2^a(0) = [0.1, 0.2, -0.1]^T$ rad, $p_3^g(0) = [0.1, 1.8]^T$ m, $p_4^g(0) = [-1.5, 0.1]^T$ m, $p_5^g(0) = [-0.1, -1]^T$ m, $p_6^g(0) = [-0.7, 1]^T$ m, $\eta_i^g(0) = 0$ rad, and all adaptive laws are set as zero initial values.

To testify the robustness of the proposed strategy, it is assumed that the actual vehicle parameters are 10% larger than the nominal values. The external disturbances are $\vartheta_{p,i}^a = (-1)^i 0.25 \sin(0.5t) \mathbf{1}_{n_i}$, $\vartheta_{\eta,i}^a = (-1)^i 0.15 \sin(0.5t) \mathbf{1}_{n_i}$, and $\vartheta_i^g = (-1)^i [0.2 \sin(0.4t), 0.4 \cos(0.3t)]^T$. And the heterogeneous UAVs-UGVs system suffers from the following sinusoidal communication noises (named as case 1): $\bar{a}_{10} = 0.5 \sin(0.5t)$, $\bar{a}_{21} = 0.5 \sin(0.4t)$, $\bar{a}_{32} = \bar{a}_{35} = 0.4 \cos(0.3t)$, $\bar{a}_{43} = -0.2 \sin(0.6t)$, $\bar{a}_{54} = 0.3 \cos(0.3t)$, and $\bar{a}_{63} = -0.5 \cos(0.7t)$.

The simulation results are provided in Figs. 5–9, where the formation tracking performance is demonstrated in Figs. 5–6, Fig. 7 presents the trajectories of errors and funnels, Fig. 8 shows the adaptive laws, and Fig. 9 depicts the smooth trajectories of Euclidean norm of control input. It is evident from these results that all the signals in the closed-loop system remain bounded. More specifically, the 3D trajectories of the heterogeneous UAVs-UGVs system are shown in Fig. 5. It can be

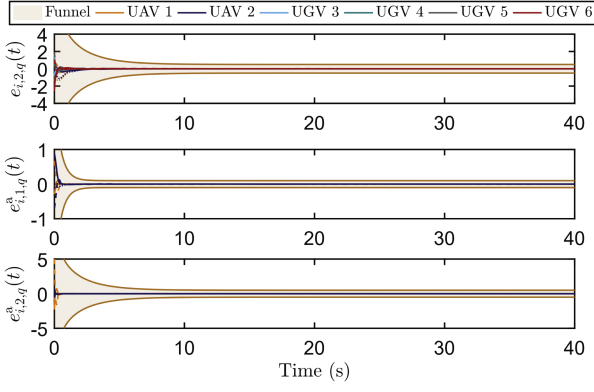


Fig. 7. Trajectories of $e_{i,2,q}$ [m], $e_{i,1,q}^a$ [m], and $e_{i,2,q}^a$ [rad].

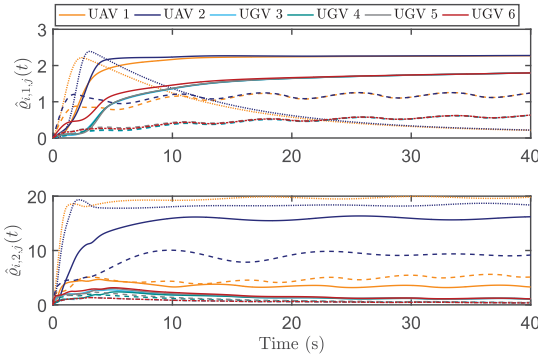


Fig. 8. Trajectories of adaptive laws: the solid line is the $\hat{E}_x$ -axis component, the dashed line is the $\hat{E}_y$ -axis component, and the dotted line represents the $\hat{E}_z$ -axis component.

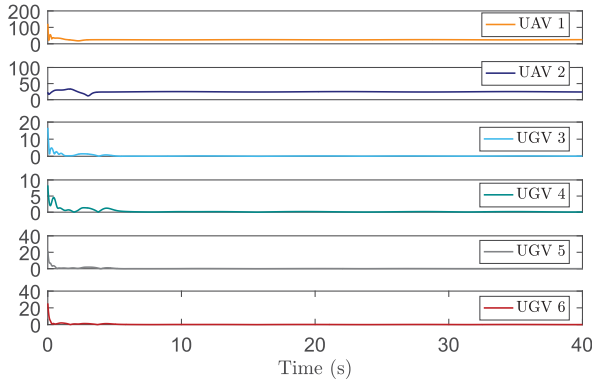


Fig. 9. Trajectories of the Euclidean norm of the control input $\|u_i\|$, $i \in \mathbf{N}_1^6$.

observed that the vehicles' trajectories track the leader's trajectory in a preset formation pattern under parametric uncertainties, disturbances, and communication noises. All errors are within the corresponding funnels and eventually converge to zero as exhibited in Figs. 6–7. Those results are consistent with the theoretical findings.

B. Comparative Study

To further demonstrate the robustness and accuracy of the proposed strategy, another type of communication noises

TABLE I
COMPARISON OF THE NORMALIZED POSITION TRACKING ERRORS UNDER TWO CASES

Index	Case 1		Case 2	
	Proposed	(50)	Proposed	(50)
$\frac{1}{n} \sum_{k=1}^n \ E_1(k)\ $	0.2237	0.5316	0.3538	0.8793
$\frac{1}{n} \sum_{k=1}^n \ E_2(k)\ $	0.3831	0.9698	0.5911	1.5103
$\frac{1}{n} \sum_{k=1}^n \ E_3(k)\ $	0.3181	0.7948	0.4088	0.9695
$\frac{1}{n} \sum_{k=1}^n \ E_4(k)\ $	0.3393	0.8155	0.4293	0.9888
$\frac{1}{n} \sum_{k=1}^n \ E_5(k)\ $	0.3644	0.8401	0.4446	1.0035
$\frac{1}{n} \sum_{k=1}^n \ E_6(k)\ $	0.3843	0.8632	0.4738	1.0347
$\frac{1}{n} \sum_{k=1}^n \ E(k)\ $	0.8838	2.0513	1.1651	2.7026

TABLE II
COMPARISON OF THE NORMALIZED CONTROL INPUTS UNDER TWO CASES

Index	Case 1		Case 2	
	Proposed	(50)	Proposed	(50)
$\frac{1}{n} \sum_{k=1}^n \ u_1(k)\ $	24.8155	24.8093	24.8110	24.8155
$\frac{1}{n} \sum_{k=1}^n \ u_2(k)\ $	24.5966	24.5468	28.4019	24.5966
$\frac{1}{n} \sum_{k=1}^n \ u_3(k)\ $	0.2311	0.2303	0.2730	0.2311
$\frac{1}{n} \sum_{k=1}^n \ u_4(k)\ $	0.1779	0.2268	0.2732	0.1779
$\frac{1}{n} \sum_{k=1}^n \ u_5(k)\ $	0.2624	0.2620	0.3315	0.2624
$\frac{1}{n} \sum_{k=1}^n \ u_6(k)\ $	0.2887	0.2664	0.3287	0.2887

(referred to as Case 2), with $\bar{a}_{10} = -0.3$, $\bar{a}_{21} = -0.2$, $\bar{a}_{32} = \bar{a}_{35} = 0.2$, $\bar{a}_{43} = -0.2$, $\bar{a}_{54} = 0.3$, and $\bar{a}_{63} = -0.3$, was introduced. Additionally, to ensure a fair comparison of the accuracy of the formation specifications, the nonlinear term of the strategy in [32] is reduced to the same levels, which can be proven using the method in [22], as follows:

$$\begin{aligned} \nu_{i,q} &= -c_{i,1} \epsilon_{i,1,q}, \quad i \in \mathbb{M}_a \cup \mathbb{M}_g, \quad q \in \mathbf{N}_1^{n_i}, \\ u_{i,q} &= -\sum_{k=1}^{n_i} [\Xi_i]_{q,k} c_{i,2} \epsilon_{i,2,k}, \end{aligned} \quad (50)$$

with the same funnel function, the same transformation as in this paper, and all other design parameters remaining the same. The normalized position tracking errors and control inputs, based on $n = 4000$ collected samples with a step size of 0.01 s, are presented in Tables I and II. Through computational analysis, the mean value of the minimum singular value of $\mathbb{H}(t)$ is found to be 0.4629 for Case 1 and 0.3824 for Case 2. According to the inequality $\|E(t)\| \leq \|e_1(t)\| / \text{svd}(\mathbb{H}(t))$, it can be deduced that the position tracking error in Case 1 converges to a tighter bound. Both tables indicate that, for all cases, the proposed strategy yields superior tracking while incurring a lower increase in control energy as compared to (50). Fig. 10 illustrates the Euclidean norm of the position tracking error trajectory, from which it is evident that the position tracking error under the proposed strategy decreases more rapidly and asymptotically converges to zero, while the position tracking error in (50) converges more slowly and ultimately stabilizes within a bounded set. This demonstrates that, under different communication noises

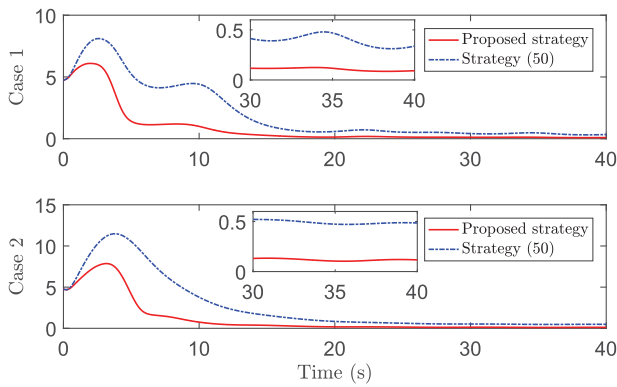


Fig. 10. Trajectories of the Euclidean norm of the position tracking error under different cases.

conditions, the proposed strategy consistently achieves superior tracking performance.

V. CONCLUSION

A new robust distributed formation control strategy has been developed by utilizing the adaptive asymptotic funnel control approach for heterogeneous UAVs-UGVs systems under uncertain environments. A model containing the dynamics of both UAVs and UGVs has been built. The proposed method achieves asymptotic formation control with predefined global funnel performance constraints, characterized by low computational and structural complexity, excluding function approximation, controller structure switching, and involving simpler nonlinear terms. In our follow-up works, we will extend the adaptive asymptotic funnel control strategy to heterogeneous UAVs-UGVs systems with unknown control directions.

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